

Week 1 — Core Vision Pipeline

Goal: Build YOLO + Depth foundation with real-time webcam inference.

Day	Key Tasks
Day 1	Environment setup, Install YOLO/OpenCV/PyTorch, Run YOLO on images
Day 2	Webcam object detection, FPS optimization, Live bounding boxes
Day 3	Modular architecture, Create detector.py, Refactor pipeline
Day 4	Depth Anything v2 setup, Generate depth maps, Color visualization
Day 5	YOLO + depth fusion, Distance estimation, Overlay labels
Day 6	Latency logging, FPS tracking, CSV metrics
Day 7	README, GitHub cleanup, Demo recording

Expected Deliverables

- Real-time YOLO webcam detection
- Depth estimation overlay
- 10–15 FPS performance
- Modular clean architecture

This week contributes directly toward building a complete real-time scene understanding research system capable of object detection, depth estimation, segmentation, narration, and intelligent perception.